AQ2.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.

Level 1:

1 point

Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}$A = $\begin{bmatrix} & & \\ & & \end{bmatrix}$. If

- $P = -1M_{11} - 2M_{12} + 0M_{13}$ P = −1M11 − 2M12 + 0M13
- $Q = -2M_{21} + 0M_{22} + 1M_{23}$ Q = −2M21 + 0M22 + 1M23
- $R = -1M_{31} - 0M_{32} + 1M_{33}$ R = −1M31 − 0M32 + 1M33

where M_{ij} Mij is the minor with respect to the (i, j)(i, j)-th entry, then choose the set of correct options.

- ☐ $det(A) = -2$det(A) = −2.
- ☐ $P \neq Q$P = Q.
- ☐ $P \neq Q \neq R$P = Q = R.
- ☐ $P = Q = R$P = Q = R.
- ☐ $Q \neq R$Q = R

No, the answer is incorrect.

Score: 0

Accepted Answers:

$det(A) = -2$det(A) = −2.

$P = Q = R$P = Q = R.

1 point

Suppose for a real $3 \times 3$3 × 3 matrix AA, there exists a real $3 \times 3$3 × 3 matrix PP such that $D = PAP^{-1}$D = PAP−1 is a real $3 \times 3$3 × 3 diagonal matrix. Choose the correct set of options.

- ☐ $det\ (A)$det (A)must be equal to $det\ (P)$det (P).
- ☐ $det\ (A)$det (A) must be equal to $det\ (D)$det (D).
- ☐ The matrix AA must be equal to DD.
- ☐ If DD is the identity matrix of order 33, then AA must be the identity matrix of order 3

No, the answer is incorrect.

Score: 0

Accepted Answers:

$det\ (A)$det (A) must be equal to $det\ (D)$det (D).

If DD is the identity matrix of order 33, then AA must be the identity matrix of order 3

1 point

Choose the set of correct options for square matrices of order 33.

☐ If AB = 0AB = 0, then one of the matrices between AA and BB must have determinant 00.

☐ If AB = 0AB = 0, then both the matrices AA and BB must have determinant 00.

☐ If AB = 3IAB = 3I, where II denotes the identity matrix of order 33, then $detA$ = $\frac{3}{det(B)}$ detA = det(B)3.

☐ If AB = 3IAB = 3I, where II denotes the identity matrix of order 33, then $detA$ = $\frac{9}{det(B)}$ detA = det(B)9.

☐ If AB = 3IAB = 3I, where II denotes the identity matrix of order 33, then $detA$ = $\frac{27}{det(B)}$ detA = det(B)27.

☐ If AB = 3IAB = 3I, where II denotes the identity matrix of order 33, then both AA and BB must have non-zero determinant.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If AB = 0AB = 0, then one of the matrices between AA and BB must have determinant 00.

If AB = 3IAB = 3I, where II denotes the identity matrix of order 33, then $detA$ = $\frac{27}{det(B)}$ detA = det(B)27.

If AB = 3IAB = 3I, where II denotes the identity matrix of order 33, then both AA and BB must have non-zero determinant.

1 point

Consider a matrix A = $\begin{bmatrix} abc \\ bca \\ cab \end{bmatrix}$A = $\begin{bmatrix} \\ abcbcacab \end{bmatrix}$. If $a + b + c$a + b + c is divisible by 66 then choose the set of correct options.

☐ $det(A)$det(A) is divisible by 55.

☐ $det(A)$det(A) is divisible by 66.

☐ $det(A)$det(A) is divisible by 22.

☐ $det(A)$det(A) is divisible by 33.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$det(A)$det(A) is divisible by 66.

$det(A)$det(A) is divisible by 22.

$det(A)$det(A) is divisible by 33.

Let A = [a_{ij}]A = [aij] be a square matrix of order 3, where a_{ij} = $i$aij = i. Find $det(A)$det(A).

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0
1 point

Level 2:

Let $A = [\delta_{ij}]$A = [δij] be a square matrix of order 33, where $\delta_{ij} = i \times j$δij = i × j. Find $det(A)$ det(A).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point
1 point
Consider the following square matrices:

$A = \begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \\ 2019 & 2020 & 2021 \end{bmatrix}$, $B = \begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$ and $C = \begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$A = $\begin{bmatrix} 2013 & 2016 & 2019 \\ 2014 & 2017 & 2020 \\ 2015 & 2020 & 2021 \end{bmatrix}$, B = $\begin{bmatrix} 2016 & 2013 & 2019 & 2017 \\ 2014 & 2020 & 2020 & 2022 \\ 2015 & 2020 & 2021 & 2021 \end{bmatrix}$ and C = $\begin{bmatrix} 4032 & 2013 & 2019 & 4034 \\ 2014 & 2020 & 2040 & 4420 \\ 1520 & 2021 & 2021 & 2021 \end{bmatrix}$

Choose the set of correct options.

- ☐ $det(B) = det(A)$det(B) = det(A).
- ☐ $det(A) = -det(B)$det(A) = −det(B).
- ☐ $det(C) \neq -2det(B)$det(C) = −2det(B).
- ☐ $det(C) = -2det(A)$det(C) = −2det(A).

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(A) = -det(B)$det(A) = −det(B).
$det(C) \neq -2det(B)$det(C) = −2det(B).
$det(C) = -2det(A)$det(C) = −2det(A).

(Based on the below information answer questions 8 and 9).

Let AA be a real $3 \times 3$3 × 3 matrix as given below:

$$A = [C_1\ C_2\ C_3]$$A = [C1C2C3]

where C_i Ci represents the ii-th column of the matrix AA. Now let us consider the following set of matrices.

- A_1 = [C_1 C_2 + 5C_3 C_3]A1 = [C1C2 + 5C3C3]
- A_2 = [C_1 + C_2 + C_3 C_2 C_3]A2 = [C1 + C2 + C3C2C3]
- A_3 = [C_1 C_2 + 5C_3 0]A3 = [C1C2 + 5C30]
- A_4 = [C_1 + C_2 C_2 + C_3 C_3 + C_1]A4 = [C1 + C2C2 + C3C3 + C1]

1 point

Which of the matrices have the same determinant as that of AA?

☐ A_1 A1

☐ A_2 A2

☐ A_3 A3

☐ A_4 A4

No, the answer is incorrect.

Score: 0

Accepted Answers:

A_1 A1

A_2 A2

1 point

Choose the set of correct options.

☐ $det(A_3)$det(A3) cannot be determined from the given information.

☐ $det(A_3)$ = 0det(A3) = 0

☐ $det(A_4)$ = 2 $det(A)$det(A4) = 2 det(A)

☐ $det(A_2)$ = 3 $det(A)$det(A2) = 3 det(A)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$det(A_3)$ = 0det(A3) = 0

$det(A_4)$ = 2 $det(A)$det(A4) = 2 det(A)

If the diagonal entries of a $3 \times 3$3 × 3 lower triangular matrix AA are 1,2, and 3, then find the sum of roots of the equation $det(A - xI)$ = 0det(A − xI) = 0, where II is the identity matrix of order 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

Check Answers

Your score is: 0/10